Designing Flawless Products with FreeCAD

FreeCAD and Python work in partnership to deliver perfect product designs. Here's how...

Computer aided design (CAD) has been around for a while to help design, develop, prototype and simulate 3D objects before they are manufactured. 3D modelling and CAD provide engineers with the look-and-feel of the product so that the required changes can be done before it's sent for production.

Common use cases of 3D CAD and 3D modelling in industry include reverse engineering, tool and mould design, stress and load analysis, CNC and CAM integration, and rapid prototyping (3D printing). Industries that use this technology include electronics and PCB design, aerospace, automotive, architecture and construction, medical devices, AR/VR development, and animation and gaming.

According to Grand View Research, the global market size for 3D modelling and 3D mapping was around 7 billion dollars in 2024 and is predicted to be more than 16 billion dollars by 2030.

A number of tools and software products are available for 3D CAD and 3D modelling (Table 1).

Why industries need 3D CAD and 3D modelling

- Faster product development
- Reduced design errors
- Easy design modification
- Better cross-team collaboration
- Cost optimisation
- Virtual testing before production
- Digital manufacturing readiness
- Improved design accuracy
- Compliance with standards
- Industry 4.0 and Industry 5.0 enablement
- Integration of customisation with AI for Industry 5.0

Using FreeCAD for 3D modelling and floorplans

URL: *https://www.freecad.org/*

FreeCAD is popular free and open software with 3D parametric modelling features for designing real world objects of any dimension and size.

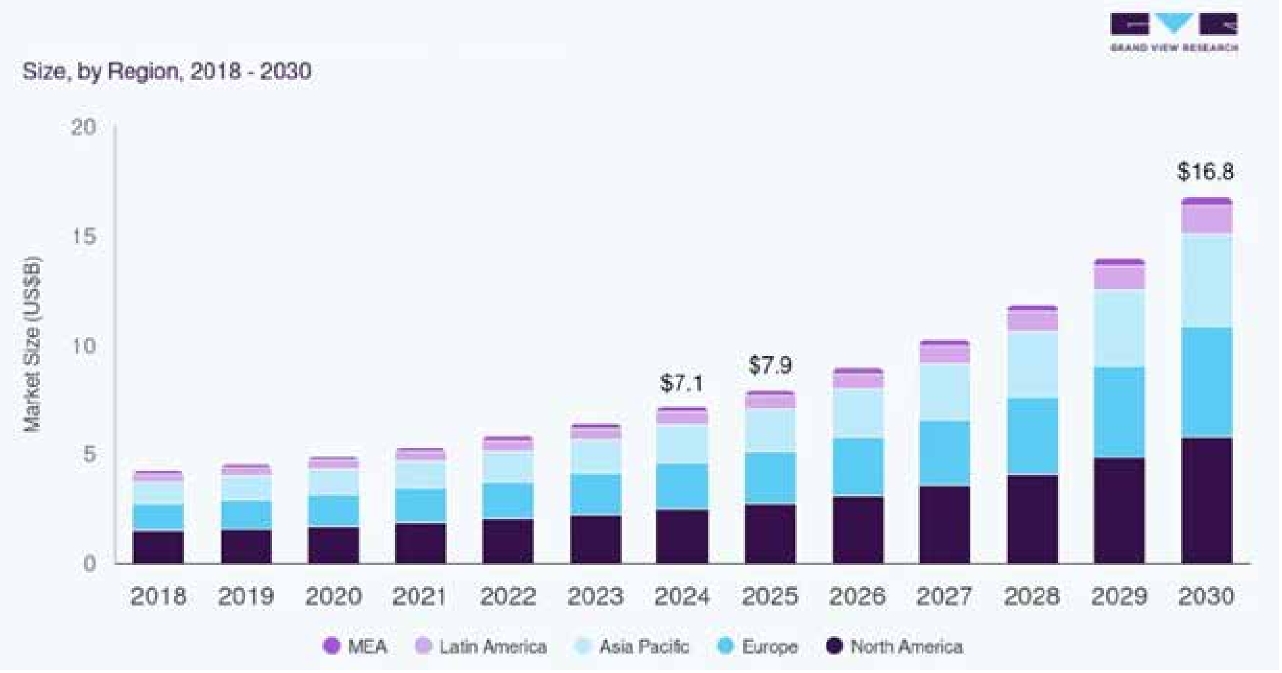

Figure 1: Market size of 3D modelling and 3D mapping